

energy - practice problems

name _____

1. A student pushes a cart with an 18 N horizontal force while the cart moves 5.0 m forward across the lab floor. How much work does the student do on the cart?

2. A box is pulled 250 cm across a smooth floor with a 40 N horizontal force. How much work does the pull do?

3. A rope pulls a sled with a 60 N force at 30° above the horizontal. The sled moves 4.0 m across level snow. How much work does the rope do?

4. A 20 kg crate slides 6.0 m across a level floor. The coefficient of kinetic friction is 0.30. How much work does friction do on the crate?

5. A person carries a 30 N grocery bag across a room at a steady pace. The bag moves 8.0 m horizontally, and the person's hand pushes up on it with 30 N. How much work does the hand do on the bag?

6. A person pushes a lawn mower with a 50 N force along its handle. The mower moves 3.0 m across the lawn. How much work does the push do?

7. A 1500 g toy car rolls across the floor at 4.0 m/s. What is its kinetic energy?

8. A 1200 kg car is moving at 54 km/h. How much kinetic energy does it have? How much kinetic energy does it have at 108 km/h, twice as fast?

9. A 4.0 kg bowling ball rolls down the lane with 50 J of kinetic energy. How fast is it moving?

10. A shot put leaves an athlete's hand at 6.0 m/s with 72 J of kinetic energy. What is the mass of the shot?

11. A 2.0 kg book falls from a shelf 1.5 m above the floor onto a desk that is 0.75 m tall. What is the change in the book's gravitational potential energy?

12. A 750 g water bottle is lifted 1.2 m from the floor onto a table. How much gravitational potential energy does it gain?

13. A 10 kg box gains 196 J of gravitational potential energy when it is lifted. How high was it lifted?

14. An 80 kg rider rolls from the top of a 2.5 m tall ramp to the bottom. What is the change in the rider's gravitational potential energy?

15. A spring with a spring constant of 180 N/m is stretched 0.20 m to the right of equilibrium. What force does the spring apply?

16. To measure a spring's constant, a student hangs different masses from it and measures how far it stretches. Use the data to find the spring constant.

mass (kg)	0.10	0.20	0.30
stretch (cm)	4.9	9.8	14.7

17. A 200 N/m spring is stretched 15 cm. How much elastic potential energy does it store? How much would it store if it were stretched twice as far?

18. A spring stores 4.0 J of energy when it is stretched. Its spring constant is 200 N/m. How far is it stretched?

19. A spring launcher does 20 J of work on a 0.50 kg ball. Assuming all of that work becomes kinetic energy, how fast does the ball leave the launcher?